1a) $\left[\begin{array}{cccc|c} \boxed{1} & 1 & 2 & 4 & 1 \\ 3 & c & 2 & 8 & c \\ 0 & 0 & 2 & 2 & 0 \end{array}\right]$

$x_c = x_p + x_s$ → Nullspace.

4 var, 3 eqn.

x_4

$\Rightarrow \left[\begin{array}{cccc|c} \boxed{1} & 1 & 2 & 4 & 1 \\ 0 & \boxed{c-3} & -4 & -4 & c \\ 0 & 0 & \boxed{2} & 2 & 0 \end{array}\right]$

$Ax = 0$

$\rightarrow c = 3$

$c \neq 3 \rightarrow$ Non zero value.

$\boxed{x_4 = 1}$

(nonzero) $(c-3)x_2 = 0$

↓ must be zero.

$\therefore \boxed{x_2 = 0}$

$2x_3 + 2x_4 = 0$

$\rightarrow \boxed{x_3 = -1}$

$x_1 - 2 + 4 = 0$

$\boxed{x_1 = -2}$

$\therefore x_s = \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

Particular

Free var = 0

$x_4 = 0$

$x_3 = 0$

$x_2 = 1$

$\hookrightarrow k \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

$x_1 + 1 + 0 = 1$

$x_1 = 0$

For $c = 3$

$$\left[\begin{array}{cccc|c} \boxed{1} & 1 & 2 & 4 & 1 \\ 0 & 0 & \boxed{-4} & -4 & 0 \\ 0 & 0 & 2 & 2 & 0 \end{array}\right]$$

need 0 here.

$$\Rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 2 & 4 & 1 \\ 0 & 0 & -4 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$k \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + l \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

2 free cols

2 soln

x_s

x_p

1b) (i) $y_1 + y_4 = y_3$

$y_1 + y_4 - y_3 = 0$

$y_2 + y_5 - y_1 = 0$

$y_3 + y_6 - y_2 = 0$

$-y_1 - y_5 - y_6 = 0$

4×6

↳ 4 eqn. 6 var.

$b \rightarrow$ solve $Ax = 0$

$$\left[\begin{array}{cccccc|c} 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & -1 & 0 \end{array}\right]$$

b) Find Nullspace.

$x_3 = 1 \mid x_3 = 0$

$x_5 = 0 \mid x_5 = 1$

$x_6 = 0 \mid x_6 = 0$

$$\left[\begin{array}{cccccc|c} 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

2a) $\rightarrow A \begin{bmatrix} a & c \\ 1 & d \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$

$2a + 5c = 2$

$2b + 5b = 5$

$\begin{bmatrix} 1 \\ \end{bmatrix} \begin{bmatrix} 2 \\ \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} \\ -6 \end{bmatrix}$$

$$a + 3c = -2$$

$$b + 3d = -6$$

$$\left[\begin{array}{cccc|c} 2 & 0 & 5 & 0 & 2 \\ 0 & 2 & 0 & 5 & 8 \\ 1 & 0 & 3 & 0 & -2 \\ 0 & 1 & 0 & 3 & -6 \end{array}\right]$$

$$A = \begin{bmatrix} 16 & -6 \\ 45 & -17 \end{bmatrix}$$

Sir's fav qstn

(2b) →

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

rank 1 rank 1 rank 1

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

rank 2

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

rank 0

(2d) Sir's fav qstn.

$u \in U$ | $u = s_1 + t_1$ | $s \in S$

$v \in U$ | $v = s_2 + t_2$ | $t_1 \in t$

$u + v$ | $s_2 \in S$ | $t_2 \in t$

$= (s_1 + t_1) + (s_2 + t_2)$

$= \underline{(s_1 + s_2)} + \underline{(t_1 + t_2)}$

$\in S$ $\in T$

closed under multiplication.

c.U

→ $c(s_1+t_1) = cs_1 + ct$

$\underbrace{c_1 s_1}_{\in S} + \underbrace{c_1 s_2}_{\in T}$

3a) $\begin{bmatrix} a & r & r & r \\ a & b & s & s \\ a & b & c & t \\ a & b & c & d \end{bmatrix}$

if $b - r \neq 0$, $c - s = 0$, $d - t = 0$

3b) A (2×2) A AB (2×2)

[A AB] → ind → Dependent

col(A)) same
col(AB))

↳ Linear comb of A.

($A B_{xy} \to A_y$)

2×4

↳ col(A) ✓